

System of Linear Algebraic Eqⁿ

① Gauss-Jacobi Method

- Applicable to the system of eqⁿ in which leading diagonal elements of the coeff-matrix are dominant (large in magnitude) in their respective rows.

Step 1:

$$\begin{aligned} |a_{11}| &> |a_{12}| + |a_{13}| \\ |a_{22}| &> |a_{21}| + |a_{23}| \\ |a_{33}| &> |a_{31}| + |a_{32}| \end{aligned}$$

Arrange in such a way that leading diagonal elements large in magnitude in respective rows satisfying above condⁿ.

Step 2: Obtain:

$$x =$$

$$y =$$

$$z =$$

Step 3:

Assuming $(x, y, z) = (x_0, y_0, z_0)$ as initial approx start iter-1.

Step 4

start iter 2 by $(x, y, z) = (x_1, y_1, z_1)$

and continue till required ans. achieved.

$$\text{Q] } \left. \begin{aligned} 10x + y - z &= 11.19 \\ -x + y + 10z &= 35.61 \\ x + 10y + z &= 28.08 \end{aligned} \right\} \text{ upto 3 decimal places.}$$

Coeff Matrix:-

$$\begin{bmatrix} 10 & 1 & -1 \\ -1 & 1 & 10 \\ 1 & 10 & 1 \end{bmatrix}$$

Here, $|a_{11}| = |10| = 10 > |a_{12}| + |a_{13}|$

$|a_{22}| = |1| = 1 > |a_{21}| + |a_{23}| = 11$

$|a_{33}| = |1| = 1 > |a_{31}| + |a_{32}| = 11$

$|a_{11}| = 10; |a_{21}| + |a_{31}| = 2; |a_{11}| > |a_{21}| + |a_{31}|$

thus system is diagonally dominant & convergence is assured.

$10x + y - z = 11.19$

$\therefore 10x = 11.19 - y + z$

$\therefore x = \frac{1}{10} [11.19 - y + z]$

$z = \frac{35.61 + x - y}{10}$

$y = \frac{28.08 - x - z}{10}$

Let initial approx. $x_0 = y_0 = z_0 = 0$.

Iter-1

$$x_1 = \frac{11.9 - y_0 + z_0}{10} = 1.1190$$

$$y_1 = \frac{28.08 - x_0 - z_0}{10} = 2.8080$$

$$z_1 = \frac{35.61 + x_0 - y_0}{10} = 3.5610$$

} putting
 $x_0 = y_0 = z_0$
 $= 0$ in
 all eqⁿ.

Iter-2

$$x_2 = \frac{11.9 - y_1 + z_1}{10} = 1.1943$$

$$y_2 = \frac{28.08 - x_1 - z_1}{10} = 2.3400$$

$$z_2 = \frac{35.61 + x_1 - y_1}{10} = 3.3921$$

} putting
 values of
 x_1, y_1 and z_1
 in all eqⁿ

Iter-3

$$x_3 = \frac{11.9 - y_2 + z_2}{10} = 1.2242$$

$$y_3 = \frac{28.08 - x_2 - z_2}{10} = 2.3494$$

$$z_3 = \frac{35.61 + x_2 - y_2}{10} = 3.4464$$

} values
 of x_2, y_2
 z_2

Iter-4

$$x_4 = \frac{11.9 - y_3 + z_3}{10} = 1.2287$$

$$y_4 = \frac{28.08 - x_3 - z_3}{10} = 2.3409$$

} and
 so on.

Continue till you get required ans...

② Gauss - Seidal Iteration Method

- Modification to Jacobi.
- Here most recent approximations are used while computing in the next step, convergence in this method is much time faster than in Jacobi method.

Step 1: Same as Jacobi

Step 2: " "

Step 3: " "

Step 4: put $x = x_1$, $z = 0$ in eqⁿ of y in $3x - 2$ and so on.

[i.e put the last modified value of the variables].

Repeat till required ans. obtained.

Q]
$$\begin{aligned} 9x + 2y + 4z &= 20 \\ 2x - 4y + 10z &= -15 \\ x + 10y + 4z &= 6 \end{aligned}$$

Coeff Matrix

$$\begin{bmatrix} 9 & 2 & 4 \\ 2 & -4 & 10 \\ 1 & 10 & 4 \end{bmatrix}$$

$|a_{11}| = 9$; $|a_{12}| + |a_{13}| = 6$; $|a_{11}| > |a_{12}| + |a_{13}|$

$$|a_{23}| = |10| = 10 > |a_{21}| + |a_{22}|$$

$$> 6$$

$$|a_{32}| = |10| = 10 > |a_{31}| + |a_{33}|$$

$$> 5$$

- System is diagonally dominant & convergence is assured.

- Let $x_0 = y_0 = z_0 = 0$.

$$x = \frac{20 - 2y - 4z}{9}$$

$$y = \frac{-15 - 2x + 4z}{10}$$

$$z = \frac{6 - x - 4y}{10}$$

Iter-1

$$\rightarrow x = x_1, y = y_0, z = z_0 \text{ for } x$$

$$\rightarrow z = z_1, y = y_1, x = x_1 \text{ for } y$$

$$\rightarrow z = z_1, x = x_1, y = y_1 \text{ for } z$$

Continue with updating values in every next eqⁿ and get required ans.